

Closed-Loop Streaming Synthesis Bounded-Drift, Arbitrary-Length Audio–Video Generation by Treating Chunk Hand-off as Feedback Control

Aleksander Majda

August 2026

Abstract

Generating arbitrarily long videos with diffusion transformers requires splitting the temporal dimension into overlapping chunks. Naive open-loop chunk-streaming accumulates exposure-bias drift: each chunk is conditioned on the previously generated latent, compounding distributional errors and causing scene collapse or grain amplification after a few hundred frames. We present **Closed-Loop Streaming Synthesis (CLSS)**, a suite of closed-loop corrections that close the feedback loop between successive chunks without modifying the transformer weights: calibrated context re-noising, per-channel EMA-tracked AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor. CLSS is implemented as a ComfyUI custom-node pipeline targeting LTX-2.3 22B (a 44 GB audio-video diffusion transformer) on consumer 16 GB VRAM hardware via GGUF 4-bit quantisation and CPU block-streaming, driven by a multi-modal guider with split per-modality classifier-free guidance and spatiotemporal guidance via a value-passthrough self-attention skip, with all streaming noise seeded for bit-reproducible runs. A two-stage pipeline generates a streaming Stage-1 draft and refines it at $2\times$ spatial resolution with a distilled schedule.

We report the production system at its operating point: single-scene, 69.3-second clips streamed as four chunks of 52 latent frames (≈ 17.3 s windows, inside the model’s ≈ 20 s positional-encoding wall), text-only (T2V) and image-to-video (I2V) variants, with the final outputs verified perceptually. Two system-level findings are validated. First, the production chunk length carries a sigma-schedule conflict: the connected scheduler derives its shift from the chunk template’s token count, and at 52-lf chunks the video requires the resulting 4.68 shift while the audio requires the 1.844 shift — no shared schedule serves both modalities, so the pipeline splits them inside the single joint pass (audio shift scaled by 0.394, shape-only, full-noise start). Second, Stage-2 audio

refinement is joined at a sigma cutoff: re-inventing audio from the Stage-2 noise level destroys it because melody and timbre are not text-specified, while a join at $\sigma = 0.55$ preserves 45% of the Stage-1 audio and adds the low- σ descent the audio would otherwise never receive. An open/closed-loop ablation at the production geometry confirms the corrections are what bound drift: without them video std rises +0.118 over five chunks, audio RMS +34%, and audio high-frequency energy drifts to $1.60\times$ its chunk-1 reference; with them video std holds to +0.001 and object identity holds 0.980–0.995 (per-cell minimum 0.79). In the verified production runs, Stage-1 video boundary similarity holds 0.93–0.99, the decoded seam-grid object identity holds 0.988–0.998 mean cell cosine, video latent std drifts < 0.005 , the audio RMS anchor holds loudness to its chunk-1 reference ($0.436 \rightarrow 0.504$ upward-only), and the Stage-2 joined refinement holds audio fidelity to Stage-1 at 0.66–0.74. Coarse scene layout is monitored by a sign-free power-map metric, and a per-chunk telemetry suite (settings dumps, per-frame drift-event tables, phase-lock detection) localises every reported failure in wall-clock time and latent-frame position.

1 Introduction

Modern video diffusion transformers operate on patchified latent representations where temporal resolution is down-scaled $8\times$ relative to pixel frames. Fitting a 30-second clip into a single forward pass requires very large GPUs. The standard remedy is *temporal chunking* via a Streaming Latent Buffer (SLB): generate the video in short overlapping segments, conditioning each on the previous denoised output. Section 2 reviews this mechanism and the exposure-bias problem it introduces.

CLSS closes the feedback loop between chunks by applying five lightweight post-denoising corrections that together bound the closed-loop gain below one, preventing drift from accumulating without modifying the transformer weights. The corrections are applied purely at the boundary between chunks and add negligible computational overhead.

Contributions.

1. Five closed-loop correction modules for temporal chunk streaming: calibrated context re-noising, EMA-tracked per-channel AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor that closes the progressive-softening channel the one-sided shrinkage leaves open (Sections 3.1–3.5).
2. A linear stability analysis of the feedback loop in latent space with an empirical per-chunk gain measurement protocol. The measurement resolves an apparent paradox: the one-step boundary sensitivity ($g_{\text{SLB}} \approx 15\text{--}87$) sits an order of magnitude above the nominal stability boundary, yet the closed loop is observably stable — the contraction back toward the data manifold is what bounds drift (Sections 3.6, 7).

3. A per-modality sigma-schedule split inside a single joint diffusion pass, resolving the measured conflict at the production chunk length where the video and the audio require different schedule shifts (Sections 5.4, 6.5).
4. A Stage-2 joined audio refinement: the audio re-denoises on the same low- σ descent as the video, joined at a sigma cutoff that keeps the unprompted content intact (Sections 5.2, 6.6).
5. A memory-efficient two-stage pipeline for LTX-2.3 22B on 16 GB VRAM using GGUF 4-bit quantisation and CPU block-streaming (Section 5).
6. A multi-modal guider for ComfyUI (split per-modality CFG, STG via value-passthrough self-attention skip) that correctly handles ComfyUI’s packed audio-video latent representation (Section 5.6).
7. ComfyUI custom nodes implementing the full pipeline with seeded, bit-reproducible streaming noise and per-chunk telemetry: unconditional settings dumps, per-frame origin/layout drift event tables, a phase-lock repetition detector, and a spatially-resolved seam identity metric (Sections 5.5, 6).
8. Production-configuration results verified perceptually: matched 69.3-second T2V and I2V generations at 52-lf chunks with the per-modality split and joined Stage-2 refinement, an open/closed-loop ablation at the same geometry, a sign-free coarse power-map layout metric, and an event-triggered seam-pin safeguard (Sections 6, 6.8).

2 Background

2.1 LTX-2.3 Architecture

LTX-2.3 is a 22-billion-parameter diffusion transformer for joint audio-video synthesis. It operates on a latent space with an $8\times$ temporal downscale (video VAE, causal) and a $160\times 4\times$ audio downscale (audio VAE, 16 kHz, hop length 160, compression factor 4), yielding a video latent rate of $25/8 = 3.125$ lf/s and an audio latent rate of $16000/(160\times 4) = 25.0$ lf/s. The model accepts interleaved video and audio tokens together with Gemma-3 text embeddings and is trained with flow-matching on the rectified-flow schedule.

The positional-encoding wall. Audio-video RoPE positions are measured in *seconds*, computed per forward pass with $\text{max_pos} = 20$ for both modalities. Any single window longer than ≈ 20 s runs off the positional grid: measured audio glitches at ≈ 20.5 s (a transient at latent frame 512–513, at the same position across seeds and resolutions) and video goes near-static. Every streaming chunk is therefore kept below the wall; the production geometry uses 52-lf windows (≈ 17.3 s new content plus overlap ≈ 2.7 s = 20.0 s total), which is the largest window that fits.

2.2 Streaming Latent Buffer (SLB)

The SLB provides temporal continuity in chunk streaming. When generating chunk N , the last overlap.lf latent frames from chunk $N - 1$ are prepended to the new chunk’s latent, and the denoiser is conditioned on them at noise level τ_c :

$$\tilde{L}_{\text{overlap}} = (1 - \tau_c) \hat{L}_{\text{overlap}} + \tau_c \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \quad (1)$$

Setting $\tau_c = 0$ yields maximal continuity but maximal drift risk; $\tau_c = 1$ decouples the chunks entirely. In LTX’s framework this is implemented as `VideoConditionByLatentIndex` with `strength = 1 - \tau_c`.

Implementation note: masked timestep, not a one-shot draw. Eq. (1) describes the target distribution of the overlap context, but the sampler does not literally draw ε once and hand the model a truncated schedule from τ_c . Tracing the actual mechanism shows that the per-token timestep supplied to the transformer at every sampling step t is $\tau_{\text{eff}} = (1 - \text{strength}) \cdot \sigma_t = \tau_c \sigma_t$ for overlap tokens, versus σ_t for the rest of the sequence — a *persistent* per-step rescaling of the global noise schedule, applied at every step, not a single noise-injection event. For a *linear* (rectified-flow) interpolation path, $x_t = (1 - t)x_0 + t\varepsilon$, the two descriptions coincide exactly: scaling every step’s effective time by a constant factor τ_c is algebraically identical to drawing x_{τ_c} once via Eq. (1) and continuing the *same* linear ODE from τ_c down to 0. LTX-2.3 is trained with a rectified-flow objective, so the equivalence holds for this model; it would not hold in general for a non-linear noise schedule.

2.3 Exposure Bias in Streaming Diffusion

Let $\hat{L}_N$ be the denoised latent of chunk N and $f_\theta(\cdot \mid \hat{L}_{N-1})$ the denoising operator conditioned on the previous chunk. In open-loop streaming:

$$\hat{L}_N = f_\theta(\hat{L}_{N-1} + \varepsilon_N) \quad (2)$$

The transformer was trained on latents drawn from the data distribution, but at inference it receives denoised latents from prior chunks, which lie off-manifold relative to its training distribution. Any distributional shift $\delta_N = \hat{L}_N - L_N^*$ (where L_N^* is the “on-manifold” latent for the same content) is in general amplified by the open-loop gain of the denoising process. Formally, if $g = \|\partial f_\theta / \partial L\|$ evaluated at L_{N-1}^* , then $\|\delta_{N+1}\| \approx g \|\delta_N\|$. When $g > 1$ this gives $\|\delta_N\| \sim g^N \rightarrow \infty$.

In practice g is not a global constant: it depends on the noise level schedule, step count, guidance strength, and the spectral content of the latent. Section 3.6 describes how g is estimated empirically per chunk via a perturbation measurement.

3 Closed-Loop Streaming Synthesis

CLSS adds five closed-loop correction modules that together bound the error accumulation by ensuring $\rho_{\text{closed}} = g \cdot (1 - \beta) \cdot (1 - \tau_c) < 1$.

3.1 Calibrated Context Re-Noising

The overlap latent fed as context to chunk N is re-noised to level τ_c via Eq. (1). This forces the denoiser to actively re-project the boundary context onto the data manifold rather than accepting it verbatim, partially correcting accumulated drift in the SLB.

In practice $\tau_c = 0.05$ provides a strong temporal constraint (boundary cosine similarity > 0.97) while allowing enough distributional repair to prevent drift accumulation. Larger values ($\tau_c = 0.20$, as suggested by some prior work) loosen the constraint too much for 22B GGUF streaming, causing visible motion discontinuities.

3.2 EMA-Tracked Per-Channel AdaIN

A slow exponential moving average (EMA) with rate $\lambda \approx 0.10$ tracks per-channel mean μ_c and standard deviation σ_c across chunks:

$$\mu_c^{(N)} = (1 - \lambda) \mu_c^{(N-1)} + \lambda \hat{\mu}_c^{(N)} \quad (3)$$

$$\sigma_c^{(N)} = (1 - \lambda) \sigma_c^{(N-1)} + \lambda \hat{\sigma}_c^{(N)} \quad (4)$$

where $\hat{\mu}_c^{(N)}, \hat{\sigma}_c^{(N)}$ are the per-channel statistics of chunk N 's new frames. After generation, the overlap region is dropped and *only the new frames* L_N^{new} (Algorithm 1, line 5) are blended toward the EMA reference via AdaIN with blend factor β :

$$\tilde{L}_N^{\text{new}} = (1 - \beta) L_N^{\text{new}} + \beta \text{AdaIN}(L_N^{\text{new}} \rightarrow \mu^{(N-1)}, \sigma^{(N-1)}) \quad (5)$$

The overlap region itself is never re-touched by AdaIN — it was already corrected as *previous* chunk's L_{N-1}^{new} before being pushed to the SLB. This suppresses fast per-chunk statistical drift while allowing slow intentional scene evolution (lighting changes, colour grading) to pass through.

Two safety caps prevent over-correction:

- **Sigma max drift:** the EMA σ_c is clamped to at most $(1 + \Delta) \times \sigma_c^{(0)}$ (default $\Delta = 0.05$), preventing AdaIN from amplifying late-chunk residual noise.
- **AdaIN max amplification:** per-channel upward scaling is capped at $1.2\times$ the current chunk's std (default), preventing AdaIN from boosting noise in channels that have become quiet.

3.3 Frequency-Band Soft Shrinkage

The latent is decomposed into $B = 4$ frequency bands via 3-D FFT along the (frame, height, width) axes. Each band b receives a continuous gain:

$$g_b^{(c)} = \min \left(1, \left(\frac{E_{\text{ref},b}^{(c)}}{E_b^{(c)}} \right)^{\gamma_b} \right) \in (0, 1] \quad (6)$$

where $E_b^{(c)}$ is the mean squared amplitude of band b in channel c , $E_{\text{ref},b}^{(c)}$ is the corresponding EMA reference, and γ_b is the per-band shrinkage exponent.

Only bands with excess energy relative to their EMA are attenuated. Bands with $\gamma_b = 0$ (the DC band) are never attenuated. Default exponents: $\gamma = (0.0, 0.05, 0.2, 0.3)$, which are conservative enough to preserve scene structure while suppressing temporal flicker.

The EMA is updated with the *corrected* (post-gain) energies, not the raw ones, so the reference tracks intentional scene content rather than accumulated error.

3.4 Dynamic Anchor Bank

A small bank of $M \leq 8$ keyframe anchors provides long-range visual identity conditioning. The bank is seeded from the *last* frame of chunk 1 (not the first) and updated when two consecutive chunks trigger a scene-change signal (cosine similarity below threshold $\rho_a = 0.78$ for $P = 2$ consecutive chunks). A *forced insertion* every F_a chunks prevents bank stagnation on visually stable sequences where the similarity threshold is never triggered. The ComfyUI node auto-derives $F_a = \max(2, \min(5, \lceil \text{num_chunks}/4 \rceil))$ ($F_a = 2$ for the four-chunk production runs).

Anchors are retrieved via cosine similarity. In the current pipeline the bank serves as the identity-telemetry reference (each chunk is compared against its nearest banked anchor per chunk); anchor *injection* into the conditioning was tested in two wirings and both produced visible artifacts (appended keyframe tokens corrupt the joint audio-video conditioning; an in-place replacement at the first new frame morphs the content back toward earlier scene content, because the redundancy filter guarantees the retrieved anchor differs from the content being continued). The bank therefore measures identity drift without correcting it; a safe anchor-conditioning mechanism is an open design problem (Section 6.8).

3.5 Two-Band Spatial Detail Anchoring

The shrinkage of Section 3.3 is deliberately one-sided (Eq. (6) only ever attenuates), and this asymmetry has a measured long-run consequence: low-band energy inflates while high bands are only ever shrunk, so the *high-frequency share* of latent energy falls monotonically — +19.6% band-0 inflation over seven chunks in one measured run, with the uncapped band EMA itself drifting upward and legitimising the excess. Nothing in the shrinkage loop can *restore* band energy, and the perceptual result is progressive detail loss.

The detail anchor closes exactly this channel. Each frame is split into a spatial low band (a 3×3 average pool over the latent grid) and its high-band residual, and the mean squared energy of each band is equalised toward a *fixed* scene-first reference (captured at chunk 1, re-baselined on scene change) with symmetric per-chunk gains:

$$g_{\text{lo}} = \left(\frac{E_{\text{lo}}^{\text{ref}}}{E_{\text{lo}}} \right)^{1/2} \text{ clamped to } [0.90, 1.10], \quad g_{\text{hi}} = \left(\frac{E_{\text{hi}}^{\text{ref}}}{E_{\text{hi}}} \right)^{1/2} \text{ clamped to } [0.90, 1.12] \quad (7)$$

Unlike the shrinkage EMA, the reference here does not track the output — drift cannot legitimise itself — and unlike the shrinkage gains, these are symmetric: lost high-band energy is restored, not merely excess removed. The per-chunk clamp bounds

the correction so a genuine scene change cannot be fought harder than $\pm 10\text{--}12\%$ per chunk, and a $\pm 0.5\%$ dead-band makes the module a strict no-op when energies already match. The same mechanism runs per-window in Stage 2, referenced there to the upsampled Stage-1 slice of the same window (Section 5.2). In the verified production runs it holds the Stage-1 high-frequency share flat end-to-end ($0.527 \rightarrow 0.527$ I2V, $0.616 \rightarrow 0.616$ T2V; Section 6).

3.6 Stability Analysis

The AdaIN blend (Section 3.2) and context re-noising (Section 3.1) together reduce the magnitude of any per-chunk latent perturbation by a factor $(1 - \beta)(1 - \tau_c)$ before it reaches the next chunk’s denoiser. This gives the *loop gain*:

$$\rho_{\text{loop}} = (1 - \beta) \cdot (1 - \tau_c) \quad (8)$$

The combined closed-loop gain, accounting for the model’s open-loop amplification g , is:

$$\rho_{\text{closed}} = g \cdot \rho_{\text{loop}} \quad (9)$$

Bounded drift requires $\rho_{\text{closed}} < 1$. With $\beta = 0.40$ and $\tau_c = 0.05$, $\rho_{\text{loop}} = 0.57$, providing $\approx 43\%$ gain reduction per chunk before accounting for g .

Why the two factors multiply. $(1 - \tau_c)$ and $(1 - \beta)$ act on disjoint tensor regions at different points in the cycle — $(1 - \tau_c)$ on the *overlap* region when it is *read* as context (Section 3.1), $(1 - \beta)$ on the *new-frame* region when it is *written* as output (Section 3.2). Each attenuator sits on the one path the error must traverse per loop iteration: a perturbation in chunk N ’s output is written with gain $(1 - \beta)$ (AdaIN is applied uniformly across L_N^{new} , including the tail frames that become the next SLB), then read back with gain $(1 - \tau_c)$ once it becomes chunk $N+1$ ’s overlap context. For a linear, time-invariant loop this is the standard construction of a loop gain from attenuators in series. What Eq. (8) does not capture is any cross-term where the correction strength itself depends on chunk content (e.g. the AdaIN amplification cap engages non-uniformly); that content-dependence is exactly why g cannot be folded into a single constant either (below).

Hyperparameter choices. $\beta = 0.40$ is the largest value that preserves scene-change responsiveness in practice: at $\beta \geq 0.5$ slow colour shifts accumulate due to excessive EMA anchoring. $\tau_c = 0.05$ is a conservative re-noising level that maintains high boundary similarity (> 0.97) while partially correcting SLB distribution shift; larger values ($\tau_c \geq 0.20$) cause visible motion discontinuities on 22B streaming. Together they yield $\rho_{\text{loop}} = 0.57$. $\lambda = 0.10$ (EMA rate) was chosen to respond to gradual scene evolution without over-weighting individual chunks; $\gamma = (0.0, 0.05, 0.2, 0.3)$ are conservative exponents that attenuate only bands with clear excess energy.

Per-chunk gain measurement. g is measured empirically by running a second denoising pass with a slightly perturbed overlap latent $\tilde{L}_{\text{overlap}} + \delta$ ($\|\delta\|_F = \epsilon \|\tilde{L}_{\text{overlap}}\|_F$, $\epsilon = 0.01$) and comparing the output perturbation to the input:

$$g_{\text{SLB}} = \frac{\|\Delta_{\text{output}}^{\text{SLB}}\|_F}{\|\delta\|_F} \quad (10)$$

where $\Delta_{\text{output}}^{\text{SLB}}$ is restricted to the last `overlap_lf` new frames—the portion that becomes the next chunk’s SLB. This measures sensitivity of the boundary output to boundary input perturbations, which is the relevant quantity for SLB-mediated drift propagation. It does not capture drift arising from other channels (e.g., prompt conditioning or attention to non-overlap frames), so g_{SLB} is a lower bound on the true open-loop gain. The measurement is exposed on the **CLSSConfig** node (`measure_g=on`, ϵ via `measure_g_epsilon`); it doubles Stage-1 compute for the continuation chunks it runs on. Measured values at both the 16-lf and the production 52-lf geometry, and their consequences for the stability model, are reported in Section 7.

4 Algorithm

Algorithm 1 summarises the per-chunk CLSS procedure.

Algorithm 1 CLSS per-chunk generation

Require: Config: $\tau_c, \beta, \lambda, \gamma$; state: SLB, EMA, anchor bank

- 1: $L_{\text{ov}} \leftarrow \text{SLB.read}()$ ▷ may be empty for chunk 0
 - 2: $\tilde{L}_{\text{ov}} \leftarrow (1 - \tau_c) L_{\text{ov}} + \tau_c \varepsilon$ ▷ Eq. (1); Sect. 3.1
 - 3: $A \leftarrow \text{anchor_bank.retrieve}(\text{top-}m \text{ by cosine sim})$ ▷ Sect. 3.4
 - 4: $L_N \leftarrow \text{Denoise}(\tilde{L}_{\text{ov}}, A, \text{prompt}_N)$
 - 5: $L_N^{\text{new}} \leftarrow L_N[\text{overlap_lf} :]$ ▷ drop overlap region
 - 6: $L_N^{\text{new}} \leftarrow \text{AdaIN_blend}(L_N^{\text{new}}, \text{EMA}, \beta)$ ▷ Sect. 3.2
 - 7: $L_N^{\text{new}} \leftarrow \text{band_soft_shrink}(L_N^{\text{new}}, \text{band_EMA}, \gamma)$ ▷ Sect. 3.3
 - 8: $L_N^{\text{new}} \leftarrow \text{detail_anchor}(L_N^{\text{new}}, E_{\text{lo/hi}}^{\text{ref}})$ ▷ Eq. (7); Sect. 3.5
 - 9: $\text{EMA.update}(L_N^{\text{new}}, \lambda)$; $\text{band_EMA.update}(L_N^{\text{new}}, \lambda)$
 - 10: $\text{anchor_bank.update}(\text{last frame of } L_N^{\text{new}})$ ▷ Sect. 3.4
 - 11: $\text{SLB.push}(L_N^{\text{new}}[-\text{overlap_lf} :])$
 - 12: **return** L_N^{new}
-

5 Implementation

5.1 Memory Budget for 16 GB VRAM

LTX-2.3 22B in BF16 occupies ≈ 44 GB. Three complementary strategies fit it within 16 GB VRAM:

1. **GGUF 4-bit quantisation** (Q4_K_S): reduces the on-disk size to ≈ 11 GB. De-quantization to BF16 at load time requires ≈ 44 GB of pinned CPU RAM.

2. **CPU block-streaming**: all transformer blocks reside in pinned CPU RAM; only a small number of blocks are resident on GPU at a time, with the next block pre-fetched asynchronously via a dedicated CUDA stream.
3. **Temporal chunking (CLSS)**: latent memory is $O(\text{overlap_lf})$ instead of $O(\text{total_frames})$. Processed chunk latents are immediately offloaded to CPU after each chunk.

Combined peak VRAM budget at the production resolution:

Component	Peak VRAM
Model blocks (BF16, streaming)	≈ 5 GB
Activations + latents (per chunk)	≈ 4 GB
VAE / text encoder (loaded/freed)	≈ 2 GB
Total peak	≈ 11 GB

5.2 Two-Stage Pipeline

Stage 1 (Open-resolution streaming). CLSS streaming generates the full video at a coarse latent resolution ($H_L \times W_L$ latent grid, 11×20 in the production runs). Each chunk runs the full 20-step LTX scheduler. CLSS corrections are applied between chunks as per Algorithm 1.

Stage 2 (Distilled refinement). Stage 1’s coarse resolution and 20-step schedule are sufficient for temporal structure and scene content but produce visible quantisation artefacts and reduced texture detail after GGUF 4-bit dequantisation. Stage 2 addresses this by upsampling and refining the output.

The Stage 1 output is spatially upsampled $2\times$ by `LTXVLatentUpsampler`, yielding a 22×40 latent grid (1408×1024 pixels). Stage 2 then refines the full video in temporal chunks using a distilled schedule with initial noise level $\sigma_0 \approx 0.909$ and a dense manual sigma sequence (0.9094, 0.725, 0.55, 0.4219, 0.25, 0; five steps). Each chunk’s new frames are seeded from the upscaled Stage 1 latent with σ_0 -scaled noise:

$$x_{\text{start}} = \sigma_0 \varepsilon + (1 - \sigma_0) L_N^{\text{S1}} \quad (11)$$

anchoring Stage 2 to the coarse temporal structure produced by Stage 1 while allowing the refinement to add local high-frequency detail. The chunk geometry is auto-selected from a VRAM token budget (production runs: `frames_per_chunk` = 42, `overlap` = 14).

Key differences from Stage 1:

- AdaIN and spectral-shrinkage corrections are *disabled*: Stage 2 refinement is structurally anchored to the S1 latent, so distributional drift correction is unnecessary.
- The detail anchor (Section 3.5) *does* run, per window, with the reference re-computed from that window’s upscaled Stage-1 slice rather than carried from

a first-window EMA: the distilled refinement systematically adds spatial high-frequency energy relative to its Stage-1 seed, and per-window re-referencing means the correction cannot accumulate.

- A global pre-generated noise field is sliced per chunk (`_SlicedNoise`) so the per-chunk noise samples tile seamlessly at the chunk boundaries.
- Audio is refined under a join-sigma rule (Section 5.2).

Stage-2 audio refinement with a join sigma. Freezing the audio through Stage 2 leaves a structural asymmetry: the video receives a dense low- σ descent (re-noised to $\sigma_0 \approx 0.909$ and re-denoised over the distilled steps) while the audio would receive none. Simply unfreezing does not work — and the reason is informative. Full refinement from $\sigma_0 \approx 0.909$ was measured to destroy the audio (`aud_refine_sim` 0.07–0.16: only 7–16% of the Stage-1 audio survives, each window re-inventing its own content), because a melody, timbre, or voice is not in the text prompt — re-inventing audio from 91% noise means losing it, whereas the video survives the same treatment because its composition *is* prompt-specified and the detail anchor re-anchors it. The fix is a *join sigma* (`audio_join_sigma`): the audio stays frozen (mask 0, audio timestep 0) through every step with $\sigma > \sigma_{\text{join}}$ and joins the descent at the first step with $\sigma \leq \sigma_{\text{join}}$, with an explicit noise-up to that σ , keeping a fraction $(1 - \sigma_{\text{join}})$ of the Stage-1 audio as starting signal. The descent is σ -matched at every step — both modalities sit at the current step’s sigma, so the failure mode of a per-modality schedule split (Section 5.4) does not arise — and it costs no extra compute, since Stage 2 already pays for these passes and the audio is already in the (masked) latent. The production configuration uses $\sigma_{\text{join}} = 0.55$ on the dense schedule (45% of the Stage-1 audio kept, three matched steps); measured results are reported in Section 6.6. A chunk-1 onset limiter (smooth $3.5\sigma \rightarrow 4\sigma \tanh$) is applied to the refined audio because the chunk-1 re-roll otherwise re-creates a violent onset transient (measured -6.08 , $\approx 11\sigma$ at the run’s std). The known remaining weakness is content seams at the Stage-2 window boundaries (`aud_bnd` 0.30–0.61 across the production runs).

5.3 Audio Streaming

Audio and video share the same temporal chunk boundaries. Audio length accounting is anchored to *pixel time*: the causal video VAE maps a chunk of `new_lf` latent frames to $(\text{new_lf} - 1) \cdot 8 + 1$ pixel frames for the first chunk of a sequence and $\text{new_lf} \cdot 8$ pixel frames for every continuation chunk. The audio rate is taken from the chunk template, $r = \text{new_af} / ((\text{new_lf} - 1) \cdot 8 + 1)$ audio frames per pixel frame, and the SLB length follows the same rate:

$$\text{overlap_af} = \text{round}(\text{overlap_lf} \cdot 8 \cdot r) \quad (12)$$

In the production geometry (52lf new, 8lf overlap) this gives `new_af` = 427 (chunk 1) / 434 (continuations), `overlap_af` = 67. The effective video overlap is additionally clamped to `new_lf`: the SLB can never carry more frames than a chunk produces, and unclamped values were measured to silently discard motion at every seam.

Continuity via reference audio and the SLB. Audio continuity is carried by two conditioning paths. The overlap-time audio of each continuation chunk carries an audio analogue of the video SLB: the overlap audio frames are seeded from the previous chunk’s tail and re-noised at τ_c , so each chunk opens with near-verbatim continuity. Alongside the SLB, a reference-audio window (`ref_audio` tokens at negative temporal RoPE) taken from the previous chunk’s pre-overlap tail provides content conditioning; its length is `ref_audio_seconds` (production value 8.0 s = 200 frames). The reference default is the audio-overlap length (67 frames $\approx$ 2.7 s). The audio context (SLB plus reference) is envelope-flattened before conditioning: each context frame is rescaled toward the window-mean RMS (gain capped at [0.6, 1.25], asymmetric so quiet frames are never over-amplified), removing the loudness arc while preserving content and timbre.

The audio SLB is re-noised on its own schedule. At the production geometry the multiplier is 1.0 (the base τ_c schedule with an audio ceiling of 0.35); at shorter chunks (16 lf) a hotter $3\times$ schedule was perceptually validated against a per-chunk loudness-gesture repetition: the near-verbatim SLB tail circulates the previous chunk’s loudness envelope, and hotter re-noising breaks the loop while the envelope-flattened context removes the arc from the conditioning.

Upward-only RMS anchoring. Un-anchored streaming audio is a divergent loop: runs were measured with RMS growing +67% and low-mid spectral bands tripling over ten chunks. A per-chunk anchor corrects the new audio’s global RMS toward an EMA target whose drift from the chunk-1 reference (computed onset-excluded) is capped at $\pm 5\%$, before the chunk’s tail is fed forward as the next reference window. The anchor direction is *upward-only*, matching the reference streaming pipeline: a chunk quieter than the capped EMA target is boosted toward it, while a louder chunk is kept raw. A hard bidirectional version was measured cutting brighter chunks back toward the chunk-1 level — a dullness-lock. The final-assembly spectral normalisation is likewise upward-only (per-bin gains floored at 1.0, RMS pin boosts only), preserving the anti-drone direction (a high-frequency decay being boosted back) while a genuine low-frequency rise passes through and is gated on the `aud_hf` trend. One caveat is explicit: the loudness runaway the upward-only law cannot suppress (RMS $\times 4$ over a 15-chunk run measured earlier) remains unhandled, so long runs gate on the `aud_rms` trend.

Boundary smoothing. At each audio chunk boundary, a linear blend over ± 2 audio latent frames is applied in latent space before VAE decoding. This eliminates click artifacts that arise from the abrupt splice between consecutive chunks, which would otherwise be visible as discontinuities in the spectrogram.

5.4 Per-Modality Sigma Schedules

The noise schedule is not a free parameter in this pipeline: ComfyUI’s `LTXVScheduler` derives its sigma shift from the token count of the latent template connected to it,

$$\sigma_{\text{shift}} = \text{tokens} \cdot \frac{1.1}{3072} + 0.583, \quad \text{tokens} = \text{template_lf} \times H \times W, \quad (13)$$

so the schedule changes whenever the chunk length does. At the 11×20 production spatial template a 16-lf chunk template yields shift 1.844 and a 52-lf template yields 4.680; the formula reproduces the logged sigma schedules to a maximum absolute error of 5×10^{-5} . This matters because the two modalities respond very differently to the schedule at the production chunk length (52lf ≈ 17.3 s windows), and the conflict was measured in both directions:

- **Shared 4.68 schedule** (13 of 20 steps at $\sigma \geq 0.9$): the video is well-formed, but the audio is under-developed — the raw Stage-1 audio latent lands at std ≈ 0.42 against 0.98–1.26 for real music through the same VAE (Section 6.7), and the decoded output measures -34 dB RMS, L/R correlation 0.9988–0.9994 (stereo collapsed to mono), ro99 bandwidth 4.4–4.7 kHz (real music: 6–12 kHz).
- **Shared flat 1.844 schedule at 52-lf chunks**: the audio is restored (raw std 0.49, validated perceptually) but the video is destroyed — it collapses to a near-static copy of the guide image (video latent std 1.99 against a normal ≈ 1.01 ; origin similarity to the guide frame $\rightarrow 0.999$ by frame 3, i.e. every frame *is* the guide image).

At 52-lf chunks the video wants the top-heavy 4.68 schedule and the audio wants the 1.844 schedule, and no shared schedule serves both. The resolution is a *per-modality sigma schedule inside the single joint pass*: the video follows the original `LTXVScheduler` sigmas byte-identical (its ancestral noise stream is untouched — the audio draws from a separate RNG stream), while the audio follows the schedule generated by Eq. (13) with σ_{shift} scaled by `mult = 1.844/4.680  $\approx 0.394$` , keeping $\sigma_0 = 1.0$: the audio still starts from full noise and only the *shape* of its schedule is remapped. Two mechanisms make the split exact rather than approximate: a `process_timestep` patch reports the audio’s true per-token sigma (`mask  $\times a_i$`) to the model, and because `calculate_denoised` converts the velocity prediction with the *global* sigma, the sampler re-derives the audio x_0 with a_i instead — mask-aware, so only fully-denoised audio tokens are converted and mask-pinned SLB tokens are handled by the mask itself. The multiplier is auto-derived (`audio_shift_mult=0`): the node divides the perceptually validated audio target shift 1.844 by the video shift it recovers from the connected sigmas. Live validation is reported in Section 6.5.

5.5 ComfyUI Node Integration

Six custom nodes implement the full pipeline inside ComfyUI:

1. **CLSSConfig**: exposes the CLSS hyperparameters worth exposing (τ_c, β , overlap); validated internals are fixed. Diagnostic levers: `shrinkage=off` zeroes γ (the spectral-shrinkage ablation arm), and `measure_g` (with `measure_g_epsilon`) enables the g_{SLB} measurement of Eq. (10).
2. **CLSSScenePrompts**: multi-scene text input via `---` delimiters; Gemma-enhanced per scene. Output is a flat CONDITIONING list with one entry per scene, proportionally assigned to chunks.
3. **CLSSStreamingSampler**: Stage 1 streaming sampler. Accepts any ComfyUI GUIDER/SAMPLER/SIGMAS/NOISE/LATENT combination. Emits an unconditional settings dump, per-chunk telemetry, end-of-run trend summaries with per-chunk localisation, per-frame origin/layout drift event tables, and a phase-lock repetition detector. Experiment levers: `open_loop=on` disables every closed-loop correction to give the naive sliding-window baseline arm; `object_identity=on` adds the spatially-resolved seam identity metric; `auto_seam_pin=on` enables the event-triggered seam pin of Section 6.8.
4. **CLSSStage2**: Stage 2 refinement sampler with consistent per-chunk noise slicing, per-window detail anchoring, and the joined audio refinement of Section 5.2.
5. **CLSSAVGuiderV2**: the multi-modal guider (Section 5.6).
6. **CLSSAVGuider**: the legacy split-CFG patch over an existing guider, retained for compatibility and inert on current ComfyUI.

All stochastic inputs are seeded: Stage-1 video *and* audio noise are pre-generated as full-sequence fields sliced per chunk (distinct yet coherent noise per chunk, bit-reproducible runs), and every fallback noise path uses a generator derived deterministically from the seed and chunk position.

5.6 Multi-Modal Guidance and the Packed-Latent Pitfall

The reference pipeline guides each modality independently:

$$\text{pred} = c + (s_{\text{cfg}} - 1)(c - u) + s_{\text{stg}}(c - p) + (s_{\text{mod}} - 1)(c - m) \quad (14)$$

per modality, followed by a rescale $\text{pred} *= r \sigma_c / \sigma_{\text{pred}} + (1 - r)$, where u is the text-unconditional prediction, m the modality-isolated prediction (both audio $\leftrightarrow$ video cross-attentions skipped), and p the STG-perturbed prediction. The STG perturbation skips *self*-attention in designated blocks (block 28 for LTX-2.3) by replacing the attention output with the raw value projection (out = $W_V x$, gating and output projection unchanged) — a value-passthrough, not a zero-out. The working configuration runs per-modality CFG 4.0 video / 7.0 audio, STG 1.0 video / 0 audio, $s_{\text{mod}} = 1$ (the modality term of Eq. (14) vanishes and its pass is skipped — three passes per step), rescale 0.7. Audio STG is off: a joint run with audio STG on deflated the audio, and every perceptually validated run used audio STG 0.

The packed-latent representation. ComfyUI’s `CFGGuider.sample()` packs nested audio-video latents into a *single flat* $[B, 1, N_v + N_a]$ tensor before sampling and unpacks only after. A guider that tries to detect the audio-video case with `isinstance(x,`

`NestedTensor`) inside `predict_noise` therefore never sees a nested tensor and falls back to plain shared CFG — and it does so silently. The guider must instead capture the per-modality shapes in `sample()` *before* packing, detect the packed $[B, 1, N]$ representation in `predict_noise`, unpack each pass output, apply Eq. (14) per modality, and repack. Two diagnostics guard against a silent regression: the guider announces the active branch and its pass count once per chunk, and it logs the STG difference norms so a zero-valued (inert) perturbation is visible immediately. A useful invariant follows from the pass count: because each guidance term costs a transformer pass, a guidance configuration that does not change per-step wall-clock time is not actually running.

6 Experiments

All runs use the production configuration of Table 1: a single prompted scene, streamed as four chunks of 52 latent frames (≈ 17.3 s each) with 8-frame overlap, assembling to 208 latent frames (69.3s at 24 fps), refined by a five-chunk Stage 2 pass at $2\times$ resolution. The telemetry cited below is the per-chunk output of the pipeline itself (settings dumps, per-chunk metrics, trend summaries, event tables); the final decoded outputs of the production I2V and T2V runs were verified perceptually by the authors.

6.1 Shared Configuration

Table 1: Production configuration shared by all runs. Guidance values are those actually active during sampling (verified by the guider’s per-chunk `passes/step` and difference-norm telemetry, Section 5.6).

Parameter	Value
Model	LTX-2.3 22B (Q4_K_S GGUF, BF16 CPU dequant)
Geometry	4 chunks $\times$ 52 lf new + 8 lf overlap = 208 lf (69.3s)
Resolution	704 $\times$ 1280 (Stage 1, 11 $\times$ 20 latent); 2 $\times$ (Stage 2)
τ_c / β / λ / Δ_σ	0.05 / 0.40 / 0.10 / 0.05
γ	(0.0, 0.05, 0.2, 0.3)
Anchor bank	threshold 0.78, persistence 2, max 8, top-2; F_a auto (=2)
Detail anchor	on (Stage 1 scene-first ref; Stage 2 per-window ref)
Video / audio guidance	CFG 4.0 / 7.0; STG 1.0 / 0; modality 1.0; rescale 0.70
STG block	28 (three passes per step)
Audio sigma split	auto: mult = 1.844/4.680 = 0.394 (shape-only, full-noise start)
Audio continuity	SLB 67 af re-noised at τ_c (ceiling 0.35); ref 200 af (8.0s), negative RoPE
Audio anchor	RMS-only, upward-only, $\pm 5\%$ EMA cap
S1 noise temporal-corr mix	0.30 (video only)
Stage 1 steps	20 (LTXVScheduler, connected template)
Stage 2	dense sigmas 0.9094/0.725/0.55/0.4219/0.25/0, audio join $\sigma = 0.55$, au
Stage 2 geometry	auto: fpc = 42, overlap 14 ($T = 208$, 22 $\times$ 40)
Seeds	Stage 1 1125899906842624; Stage 2 809542877813341
Diagnostics	<code>object_identity=on</code> ; <code>open_loop/auto_seam_pin</code> per experiment
VRAM	15.61 GB GPU; peak usage within budget
T2V first frame	text only
I2V first frame	guide image, latent [1, 128, 1, 11, 20]

6.2 Closed-Loop versus Open-Loop Streaming

The exposure-bias account (Section 2) predicts that removing the corrections at the production geometry re-opens drift. Two same-seed I2V runs test this: `open_loop=on` (every closed-loop correction disabled: re-noising, AdaIN, shrinkage, detail and std anchors, audio RMS anchor) over five chunks (86.7s), and the closed-loop arm over four chunks (69.3s), both with `measure_g=on`.

Without corrections, drift returns exactly as predicted: video std 1.015 $\rightarrow$ 1.133 (+0.118 over five chunks), audio RMS 0.376 $\rightarrow$ 0.504 (+34%), audio high-frequency energy drifting to 1.60 $\times$ the chunk-1 reference (chunk-5 per-bin ratios up to 2.09), and the loudness-envelope correlation between consecutive chunks rising $-0.05 \rightarrow +0.59$. The closed-loop arm holds the lines the open arm loses: video std +0.001, object identity 0.980–0.995 (per-cell minimum 0.79 at the chunk-2 seam), audio high-frequency drift inside a 0.22 range. Audio RMS rises +0.087 — partly by design:

the upward-only anchor lets louder chunks keep their gain. The measured open-loop gain (Eq. (10)) reads $g_{\text{SLB}} = 24.8/28.8/42.4$ on the closed arm (chunks 2–4) and $33.8/35.9/38.4/21.4$ on the open arm (chunks 2–5) — the same 15–87 range measured at the shorter 16-lf geometry, so the transformer’s boundary sensitivity does not change with chunk length. The arms share seed and configuration but differ in length (4 vs. 5 chunks), and same-seed runs at different total lengths do not share a prefix (the pre-generated noise layout depends on total length): close to controlled, not perfectly.

6.3 Image-to-Video Generation (69.3 s)

The I2V run (guide image, four 52-lf chunks) is the reference point of the paper; its decoded output was verified perceptually.

Stage 1 video. Video latent std holds $1.011 \rightarrow 1.006$ ($\Delta = 0.004$). Boundary similarity across the three seams reads $0.949/0.977/0.986$; intra-chunk similarity grows $0.692 \rightarrow 0.910$; identity to the anchor bank holds $0.949 \rightarrow 0.986$; the decoded seam-grid object identity (8×8 cell cosine) holds mean $0.988/0.996/0.998$ with per-cell minima $0.875/0.963/0.988$. The detail anchor holds the high-frequency energy share flat end-to-end ($0.527 \rightarrow 0.527$). Origin similarity to the scene-first frame walks $0.378 \rightarrow 0.331$ — the scene moving away from its opening, not a drift failure.

Stage 1 audio. Raw audio latent std per chunk reads $0.440/0.400/0.457/0.495$ — the split schedule (Section 5.4) restoring the audio scale. The RMS anchor (upward-only) holds loudness to the chunk-1 reference: $0.435 \rightarrow 0.504$ over the run, with two chunks kept raw (louder, by design) and one boosted ($g = 1.09$). The audio SLB is honoured exactly on every continuation chunk (1.0) despite the re-noising; the loudness-envelope correlation across seams reads $0.032 \rightarrow 0.477 \rightarrow 0.808$ — the climb at this chunk length is a property of the continuity context’s shared envelope, and the output was verified free of audible repetition. Audio boundary similarity reads $0.624/0.295/0.357$ (the chunk-3 seam is the run’s weakest audio point); audio high-frequency share rises $0.775 \rightarrow 1.067$ inside the stability tolerance.

Stage 2. Five windows (auto geometry, Table 1). Video fidelity to the Stage-1 latent holds $0.981 \rightarrow 0.893$ (first frame) and $0.958 \rightarrow 0.882$ (last frame) — the known late-run softening. The joined audio refinement (Section 5.2) holds `aud_refine_sim` $0.66/0.67/0.69/0.73/0.74$ across the windows and leaves the final audio at std 0.517 against Stage-1’s 0.459: brighter, not re-invented. The audio seams between Stage-2 windows read $0.43/0.39/0.44/0.61$ — the joined descent’s known residual.

6.4 Text-to-Video Generation (69.3 s)

The T2V run (text only, same seed and configuration; the seam-pin safeguard was enabled and did not trigger) was likewise verified perceptually.

Stage 1 video. Video latent std holds $1.036 \rightarrow 1.038$ ($\Delta + 0.002$). Boundary similarity reads $0.929/0.829/0.978$; intra-chunk similarity grows $0.784 \rightarrow 0.883$; anchor identity holds $0.929 \rightarrow 0.978$; seam-grid object identity holds $0.997 \rightarrow 0.994$. The detail anchor holds the high-frequency share flat ($0.616 \rightarrow 0.616$). Origin similarity walks $0.612 \rightarrow 0.476$ (flagged drift) — without a guide image the text-only scene moves further from its opening than the I2V run does, while the anchored identity metrics stay put.

Stage 1 audio. RMS $0.446 \rightarrow 0.432$ against the chunk-1 reference; the audio SLB honoured exactly on every continuation chunk; loudness-envelope correlation across seams reads $-0.096 \rightarrow 0.555$, below the repetition flag at every seam. Audio boundary similarity reads $0.361/0.246/0.198$; high-frequency share falls $1.198 \rightarrow 1.074$.

Stage 2. Video fidelity to Stage-1 holds $0.921 \rightarrow 0.826$ (first) and $0.920 \rightarrow 0.828$ (last). Joined audio refinement holds `aud_refine_sim` $0.58/0.65/0.64/0.67/0.65$; audio seams between Stage-2 windows read $0.53/0.52/0.39/0.45$.

6.5 The Per-Modality Schedule Split

The split of Section 5.4 was validated before it became the production default (I2V, $2 \times 52\text{lf}$): the video runs the byte-identical 4.68-shift sigmas and the audio its own $0.39\times$ -shifted schedule ($\sigma_{\text{shift}} \approx 1.83$, full-noise start), `audio_stg`= 0. The audio recovers the scale the shared 4.68 schedule suppressed — raw Stage-1 audio latent std 0.497 against 0.42 on the shared schedule — while the video is untouched: video std $1.013 \rightarrow 1.008$, boundary similarity 0.967 , identity 0.967 , all in the normal range on the unchanged video sigmas. Stage-2 audio refinement (join sigma) holds `aud_refine_sim` $0.70\text{--}0.72$, and the final audio std is 0.540 . The output was validated perceptually. In the production runs the multiplier is auto-derived (`audio_shift_mult`= 0): $1.844/4.680 = 0.394$, applied shape-only with a full-noise start. An earlier joint run of the same split deflated the audio (std 0.30), but that run also carried `audio_stg`= 1 on the guider — every perceptually validated run used `audio_stg`= 0 — so the deflation is attributed to the guidance confound, not the split.

6.6 Joined Stage-2 Audio Refinement

The join-sigma design of Section 5.2 is the outcome of a measured escalation. Full refinement from $\sigma_0 = 0.909$ destroyed the audio (`aud_refine_sim` $0.07\text{--}0.16$): a melody, timbre, or voice is not in the text prompt, so re-inventing it from 91% noise loses it. Intermediate joins trace the boundary between the regimes: a join at 0.725 (stock three-step sigmas) held $0.45\text{--}0.49$; a join at 0.422 kept 58% of the Stage-1 audio and read high fidelity but without the brightness gain (rejected); the production dense schedule with $\sigma_{\text{join}} = 0.55$ keeps 45% of the Stage-1 audio as starting signal over three matched steps and holds `aud_refine_sim` $0.66\text{--}0.74$ in the I2V run (Section 6.3) and $0.58\text{--}0.67$ in the T2V run (Section 6.4), brightening the final audio

(std 0.459 $\rightarrow$ 0.517, I2V). The chunk-1 onset limiter is load-bearing: without it the chunk-1 re-roll re-creates a measured -6.08 ($\approx 11\sigma$) onset transient. The rule the escalation settled: audio is either generated in one pass, or re-denoised only from a noise level low enough that its content survives.

6.7 Audio Latent Scale and the Measurement Probe

The audio VAE’s latent normaliser is calibrated so that *real* audio lands at unit variance: encoding four real music tracks gives an overall latent std of 0.98–1.26 with the per-frequency-bin std flat at 1.00 (0.94–1.07 across all 16 bins). Generated audio does not get there, and the shortfall predicts the perceptual failure: the half-scale latents decode to exactly what is heard — quiet, low-passed near 3–4 kHz against the 8 kHz mel ceiling, and, at the longest windows, stereo collapsed to identical channels.

Table 2: Latent statistics of real and generated audio through the same VAE (probe: `simulations/audio_latent_probe.py`). ro99 is the 99th percentile frequency; L/R is the inter-channel correlation of the latent. Real music sits at unit variance; generated audio sits below it, and the shortfall tracks the audible deficits.

Source	latent std	ro99 (kHz)	L/R corr
Real music (4 tracks)	0.98–1.26	8–12	0.69–0.87
Generated, 26 s per-chunk	0.84	6.1	0.93
Generated, 20 s per-chunk	0.94	4.0	0.98
Generated, single 106 s window	0.63	3.7	0.999
Generated, single 104 s window	0.48	3.1	1.00000

The trade the table exposes: the single-window soundtrack draft buys global planning but the two longest-window runs are also the two most mean-regressed, while the strongest runs on these measures came from per-chunk windows — a mid-length window is the untested middle ground. Contrast is *not* the deficit: the draft’s 0.25 s envelope range (8.1 dB) already exceeds real music’s (4.3–6.3 dB) and its log-mel contrast (1.35) exceeds real music’s (1.00); only level and treble are short. The VAE itself is innocent: real music round-trips through it with bandwidth and stereo intact.

The probe is a small CPU script (audio VAE only, no transformer — it runs while the GPU is busy) reporting latent std, envelope dynamic range (dynR), L/R correlation, and ro99 bandwidth. Of these, dynR is the one perceptually validated axis: in a controlled A/B a 19.7 s window at dynR 9.8 dB was judged much better than a 104.3 s window at 3.3 dB (a drone), while bandwidth did *not* separate the two — bandwidth and level are not quality, and optimising them is not optimising what is heard. A post-hoc statistical repair confirms this: shifting the audio latent’s per-(channel,bin) mean toward a real-music profile improved every metric being watched (ro99 3141 $\rightarrow$ 4477 Hz, level +4 dB) and the live run was noise — the recovered “bandwidth” was a broadband noise bed and the loudness envelope collapsed. Latent

metrics localise failures; they never prove a quality win — the ear is the only ground truth.

6.8 Layout Stability and the Seam-Pin Safeguard

Coarse scene layout is monitored per frame by a sign-free power-map metric: the channel-mean spatial map of each frame is squared, hard low-passed (3×3 , stride 3), and compared against the scene-first frame by cosine similarity. Because the maps are non-negative, the metric cannot flip sign on fine latent phase, and the hard low-pass retains only the spatial scale that survives decoding. In the verified production outputs no mid-run layout events are observed: layout similarity fluctuates smoothly within its scene-normal range, and the sudden single-frame re-layouts the metric exists to catch (visible as jumping or morphing content) do not occur in the runs reported here.

As a safeguard against such events, the sampler exposes an event-triggered seam pin (`auto_seam_pin`): when a chunk’s layout series crosses the event thresholds (maximum per-frame drop > 0.25 or minimum < 0.05), the *next* chunk’s overlap is placed frozen ($\tau_c = 0$), so generation continues from the last good frames exactly — the fresh SLB, 0.3s old, rather than the anchor bank, whose entries are one to two chunks old. The trigger is evaluated from chunk 2 onward (chunk 1 is scene establishment) and cannot protect the final chunk. The mechanism localises events and anchors the continuation to the most recent frames; because the runs reported here contain no qualifying event, its behaviour against a genuine re-layout is left to future measurement, together with the underlying question of how to stop a joint audio-video model from re-laying-out the opening frames of a new chunk — the same design problem that blocks anchor injection (Section 3.4): conditioning on a temporally-displaced keyframe without corrupting audio or morphing video has no known safe wiring.

6.9 Computational Overhead

The corrections themselves add negligible compute: AdaIN, shrinkage, the detail anchor, and the boundary bookkeeping operate on chunk-sized tensors between denoising runs. The dominant cost drivers are structural. Stage 1 runs three transformer passes per step (conditional, unconditional, STG-perturbed); measured at the 16-lf geometry, chunk 1 costs ≈ 20.6 s/it and continuation chunks (carrying the 8-frame overlap) ≈ 26.8 – 33.7 s/it for a ≈ 53 -second clip, totalling ≈ 116 minutes including the ≈ 21 -minute Stage 2. The per-modality audio split, the joined Stage-2 refinement, and the `object_identity` decoding add no extra denoising passes; only `measure_g` does, doubling Stage-1 compute on the chunks it measures. In Stage 2 the per-window cost rises $\approx 50\%$ after the first window against a $\approx 32\%$ token-count rise: as the run proceeds an increasing share of the model is pushed to CPU, so the slowdown is an offload-pressure effect rather than a compute-scaling one. The production 52-lf geometry trades fewer seams (four chunks instead of ten) for larger windows; the sampler logs a startup warning whenever the connected template’s token count pushes the

schedule outside the validated shift range, so a silently changed schedule cannot go unnoticed.

7 Discussion

7.1 What the Closed Loop Bounds, and What It Does Not

The open/closed ablation (Section 6.2) is the cleanest statement of the loop’s value: at the production geometry, every correction off re-opens drift on every measured channel (video std +0.118, audio RMS +34%, audio HF $\rightarrow 1.60\times$ reference, envelope correlation climbing), and every correction on holds video std to +0.001 with object identity 0.980–0.995. What the loop does *not* bound, at either modality: the loudness-envelope correlation between consecutive chunks still climbs at 52-lf chunks (0.808 by chunk 4 in the production I2V run, below the flag in T2V), and the audio boundary metric floors mid-run in both production runs. The output-level verification matters here: the production I2V run was verified free of audible repetition despite the envelope climb — at this chunk length the climb tracks the continuity context’s shared envelope rather than a replayed gesture.

7.2 The Stability Model as a Local Design Guide

The measured boundary sensitivity $g_{\text{SLB}} \approx 15\text{--}87$ (16-lf runs) and 21–42 (52-lf runs) sits an order of magnitude above the nominal stability boundary — the naive product $\rho_{\text{loop}} \times g_{\text{SLB}} \approx 0.57 \times 26.7 \approx 15$ would predict divergence, yet the closed-loop runs are observably stable (video std $\Delta < 0.005$, boundary similarity rising or flat). The linear gain model of Section 3.6 therefore does not describe the closed loop asymptotically: the one-step sensitivity includes legitimate content re-interpretation of a 1% boundary perturbation, which the next denoise pulls back toward the data manifold, and that contraction — absent from the model — is what actually bounds drift. The model remains useful as a local design guide: which correction attenuates which channel, and by how much (the product form of Eq. (8) is validated by the mechanism, not by the numbers). A Lyapunov-style treatment or a Monte Carlo sweep over perturbation seeds and chunk positions would be needed for a rigorous account.

7.3 Object Identity at the Seam

The pooled identity metrics (mean cosine over the latent) are object-blind: a vehicle or person can be replaced across a chunk boundary while the pooled cosines stay above 0.95. The `object_identity` metric closes the observation gap by decoding the two seam frames and reporting a spatially-resolved 8×8 grid-cell cosine, so a regional content swap drops the minimum cell score even when the global mean stays high. In both production runs the metric corroborates the pooled telemetry: means 0.988–0.998 with per-cell minima ≥ 0.875 (I2V) and ≥ 0.798 -era pooled floors absent from the grid metric. The mitigation for object-critical content remains textual —

one prompt per chunk via **CLSSScenePrompts**, re-describing the persistent objects explicitly.

7.4 Scope and Generality

Single configuration, single seed, single hardware. The production results share one set of hyperparameters, one seed, and one hardware configuration (16 GB VRAM); the T2V and I2V arms differ only in the first-frame conditioning, by design. Every single-run comparison is therefore an $n = 1$ -per-arm observation, not a statistically powered result. The ablation arms and the open-loop baseline are fully specified and exposed as node levers (Section 6).

Stability metrics vs. perceptual quality. The reported metrics — boundary cosine similarity, intra-chunk similarity, identity and object-identity cosines, latent std and band energies — measure the *stability* of the latent representation across chunks. They do not assess perceptual quality, prompt adherence, or realism; the final outputs of the production runs were verified perceptually by the authors, and every audio intervention in this paper was gated perceptually (Section 6.7).

Hardware specificity. The GGUF 4-bit quantisation and CPU block-streaming configuration was designed for a specific VRAM budget (16 GB). Performance, latency, and numerical behaviour may differ on GPUs with more VRAM (where block-streaming is unnecessary) or with different quantisation schemes.

8 Related Work

Long video generation. StreamingT2V [3] and FIFO-Diffusion [4] generate long videos by sliding-window denoising. Gen-L-Video [5] uses temporal attention patching. FreeNoise [1] reschedules the initial noise (window-based fusion of a shuffled, reused noise sequence) to extend a pretrained model’s temporal attention window without retraining or sliding-window denoising at all — a tuning-free approach to the same underlying problem (fixed training-time context length) that CLSS addresses by chunking with closed-loop correction instead. VidToMe [2] merges self-attention tokens across frames to improve temporal consistency in zero-shot video editing, operating within a single generation pass rather than across autoregressively-generated chunks. CLSS differs from all of these by applying closed-loop corrections specifically designed for the latent-streaming regime of large audio-video models, where each chunk is a genuinely separate denoising process conditioned on the previous chunk’s output — the exposure-bias setting FreeNoise and VidToMe do not target.

Exposure bias in diffusion. The exposure bias problem in autoregressive models is well-studied [6]. For diffusion models, Ning et al. [7] and Salimans & Ho [8] address related distributional mismatch issues during training, but inference-time corrections for streaming have received less attention.

Statistical normalisation for consistency. Adaptive Instance Normalisation (AdaIN) [9] was introduced for style transfer, where the goal is to match the mean and standard deviation of a content feature map to those of a style reference. In CLSS, AdaIN is repurposed as a *drift correction* applied to denoised latents between chunks, using the cross-chunk EMA as the style reference. The combination of EMA tracking with AdaIN blending (rather than full replacement) is motivated by the streaming setting, where the reference must itself evolve to allow intentional scene changes.

Frequency-domain video editing. Yang et al. [10] and Ceylan et al. [11] use frequency-domain constraints for temporal consistency in video editing. CLSS applies frequency shrinkage in the *latent* domain to a streaming generation setting, targeting energy accumulation rather than pixel-domain flicker.

9 Conclusion

We presented CLSS, a five-component closed-loop streaming synthesis framework that enables stable long audio-video generation with LTX-2.3 22B on consumer 16 GB VRAM hardware. The corrections — calibrated context re-noising, EMA-tracked AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor — are applied between chunks without touching the transformer weights. At the production operating point (four 52-lf chunks, ≈ 17.3 s windows inside the model’s 20-second positional wall), the pipeline holds video latent std to $\Delta < 0.005$, seam boundary similarity at 0.93–0.99, and the high-frequency energy share flat end-to-end in both conditioning modes, while the open/closed-loop ablation shows the corrections are exactly what stands between that stability and open-loop drift (+0.118 std, +34% audio RMS, $1.60\times$ audio high-frequency runaway over five chunks).

Two system-level problems were resolved by measured design rather than by tuning. The sigma-schedule conflict — the video requires the scheduler’s 4.68 shift at 52-lf chunks, the audio the 1.844 shift — is resolved by splitting the schedule per modality inside the single joint pass (shape-only, full-noise start, auto-derived 0.394 multiplier), restoring the audio’s latent scale without touching the video’s byte-identical sigmas. The Stage-2 refinement asymmetry is resolved by the join sigma: audio joins the video’s low- σ descent at $\sigma = 0.55$, keeping 45% of the Stage-1 audio and holding its fidelity at 0.66–0.74, where full re-invention destroys it (0.07–0.16) because melody and timbre are not text-specified. The audio-scale probe (real music unit variance vs. generated half-scale, dynR as the one perceptually validated axis) gates every audio intervention, and the pipeline’s per-chunk telemetry — settings dumps, per-frame drift-event tables, phase-lock detection, seam-grid object identity — localises every failure mode reported here in wall-clock time and latent-frame position.

Two open problems remain. First, coarse-layout jump events: the layout metric and the seam-pin safeguard that target them are in place (Section 6.8), but their behaviour against a genuine event is unmeasured, since none occurs in the runs reported here. Second, object-level identity persistence: the pooled metrics are object-blind,

and the decoded seam-grid metric observes object swaps without correcting them. Both are active directions.

Acknowledgements

We thank the LTX-Video team at Lightricks for releasing LTX-2.3 and the associated GGUF weights, and the ComfyUI community for infrastructure.

References

- [1] H. Qiu, M. Xia, Y. Zhang, Y. He, X. Wang, Y. Shan, and Z. Liu. FreeNoise: Tuning-free longer video diffusion via noise rescheduling. In *ICLR*, 2024.
- [2] X. Li, C. Ma, X. Yang, and M.-H. Yang. VidToMe: Video token merging for zero-shot video editing. In *CVPR*, 2024.
- [3] R. Henschel, L. Khachatryan, D. Hayrapetyan, H. Poghosyan, R. Abrahamyan, A. Nersisyan, S. I. Yerosyan, and O. Kacher. StreamingT2V: Consistent, dynamic, and extendable long video generation from text. *arXiv:2403.14773*, 2024.
- [4] J. Kim, J. Jeong, S. Park, M. Kim, S. Kim, and J. Ye. FIFO-Diffusion: Generating infinite videos from text without training. *arXiv:2405.11473*, 2024.
- [5] F. Wang, W. Chen, G. Song, H. Zhang, W. Li, C. Shen, and T. Mei. Gen-L-Video: Multi-text conditioned long video generation via temporal co-denoising. *arXiv:2305.18264*, 2023.
- [6] S. Bengio, O. Vinyals, N. Jaitly, and N. Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *NeurIPS*, 2015.
- [7] M. Ning, M. Li, J. Su, A. A. Salah, and I. Ertugrul. Elucidating the exposure bias in diffusion models. *arXiv:2308.15321*, 2023.
- [8] T. Salimans and J. Ho. Progressive distillation for fast sampling of diffusion models. In *ICLR*, 2022.
- [9] X. Huang and S. Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *ICCV*, 2017.
- [10] S. Yang, Y. Zhou, Z. Liu, and C. C. Loy. FRESCO: Spatial-temporal correspondence for zero-shot video translation. *arXiv:2403.12962*, 2024.
- [11] D. Ceylan, C.-H. P. Huang, and N. J. Mitra. Pix2video: Video editing using image diffusion. In *ICCV*, 2023.